

Deep Learning for Molecular Binding: GNN Approach

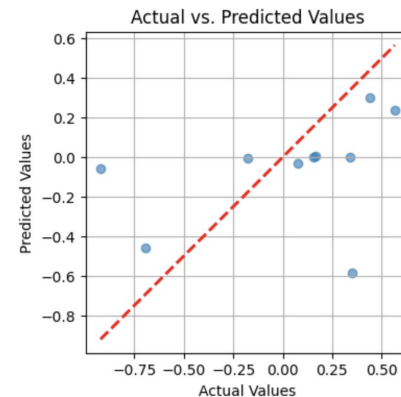
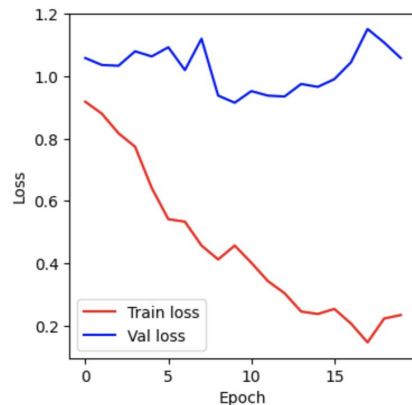
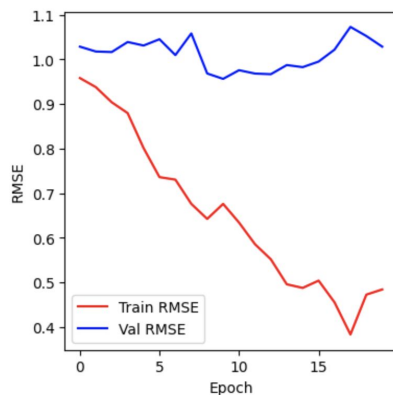
LAIDD Team2 Hyojin Kim

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The top-performing result

- Training targets = 'Autodock-Binding_Affinity(kcal/mol)', 'CNN_affinity'
- Batch size: 100
- Learning rate(lr) = 0.0001
- training/validation/test set= 6:2:2

- Edge weight: true
- Epoch#: 20
- Result: [TEST] RMSE: 0.4487
[TEST] MAE: 0.3436
[TEST] Loss: 0.20129



Model No.	1	2	3	4	5	6
lr	0.001	0.0001 (fixed)	0.0001	0.0001	0.0001	0.0001
Edge weight	true	true	true	true	true	true
layer	3	3	2	3	3	3
Test(shuffle)	false	false	false	true	false	false
Training:Validat ion:Test	8:1:1	8:1:1	8:1:1	8:1:1	6:2:2	6:2:2
Epoch#	20	20	20	20	20	10
Result	[TEST] RMSE: 1.2019 [TEST] MAE: 1.1181 [TEST] Loss: 1.44465	[TEST] RMSE: 0.7474 [TEST] MAE: 0.6508 [TEST] Loss: 0.55864	[TEST] RMSE: 1.4500 [TEST] MAE: 1.3134 [TEST] Loss: 2.10248	[TEST] RMSE: 1.3783 [TEST] MAE: 1.1568 [TEST] Loss: 1.89985	[TEST] RMSE: 0.4487 [TEST] MAE: 0.3436 [TEST] Loss: 0.20129	[TEST] RMSE: 0.9268 [TEST] MAE: 0.7259 [TEST] Loss: 0.85888
Graph	